🧠 What is NumPy?

NumPy stands for **Numerical Python**.
It's a **Python library** used for working with:

- **Numbers** (mathematical computations)
- **Arrays** (instead of lists)
- **Matrices** (multi-dimensional data)
- **Linear algebra**, **statistics**, and **random numbers**

Think of it as:

💡 "A faster, more powerful version of lists — designed for math and data analysis."

⚙️ Why NumPy?

Python lists are **flexible** but **slow** for heavy mathematical work.

NumPy provides:
✅ **Speed** — it's implemented in C
✅ **Efficiency** — uses less memory
✅ **Convenience** — supports many mathematical operations directly
✅ **Compatibility** — works with Pandas, TensorFlow, Scikit-learn, etc.

🧩 Installing NumPy

To install NumPy, run this command in your terminal:

```
pip install numpy
```

🔢 Importing NumPy

Conventionally, we import it using an alias np:

```
import numpy as np
```

🧱 Creating NumPy Arrays

The **main object** in NumPy is the ndarray (N-dimensional array).

Example:

```
import numpy as np


arr = np.array([1, 2, 3, 4, 5])

print(arr)
```

Output:

```
[1 2 3 4 5]
```

🧮 List vs NumPy Array

Feature	Python List	NumPy Array
Data Type	Mixed	Same type
Speed	Slow	Fast
Memory	More	Less
Math Ops	Manual loops	Vectorized (built-in)

Example:

```
list1 = [1, 2, 3]

arr1 = np.array([1, 2, 3])


print(list1 * 2)   # [1, 2, 3, 1, 2, 3]

print(arr1 * 2)    # [2 4 6]
```

🧭 Dimensions in NumPy Arrays

NumPy supports multiple dimensions:

```
# 0D (Scalar)
```

```
a = np.array(5)


# 1D (Vector)
b = np.array([1, 2, 3])


# 2D (Matrix)
c = np.array([[1, 2, 3], [4, 5, 6]])


# 3D (Tensor)
d = np.array([[[1,2],[3,4]], [[5,6],[7,8]]])


print(a.ndim)  # Number of dimensions
```

🧰 Basic Array Attributes

```
arr = np.array([[1, 2, 3], [4, 5, 6]])


print(arr.ndim)   # 2 (dimensions)
print(arr.shape)  # (2, 3)
print(arr.size)   # 6 elements
print(arr.dtype)  # int32 or int64
```

🧮 Common Array Operations

```
a = np.array([10, 20, 30])
b = np.array([1, 2, 3])


print(a + b)  # [11 22 33]
```

```
print(a - b)  # [ 9 18 27]
print(a * b)  # [10 40 90]
print(a / b)  # [10. 10. 10.]
```

🎯 Why NumPy is Important

NumPy is **the foundation** for:

- **Data Analysis** → Pandas uses it
- **Machine Learning** → Scikit-learn, TensorFlow
- **Data Visualization** → Matplotlib
- **Scientific Computing** → SciPy, simulations, matrix algebra

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🏁 Quick Recap

Concept	Description
Library name	NumPy
Core Object	ndarray
Advantage	Fast, memory efficient
Used in	ML, Data Science, Engineering
Built with	C (for performance)